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	Exposure of crewmembers to space-related radiation can be a serious impediment to future space travel of long duration. There is a need to understand the effects of radiation on the immune system and the mechanism(s) of radiation-induced cell damage. In particular, we will study the effect of proton, heavy ion and gamma radiation on human dendritic cells (DCs) with respect to their ability to present antigen and its impact to the immune system. The period of performance for the work proposed is for 3 years, beginning June 1, 2002 through May 31, 2005. The recruitment of peripheral blood progenitor cells and the expansion of dendritic cells with hematopoietic growth factors has been used for reconstitution of the immune system. Dendritic cells are the most potent antigen presenting cells of the immune system and are required for optimal immune activation. Our proposal is to investigate the effects of radiation on monocyte-derived DC (MMDC) that will lead to a better understanding of the mechanism(s) involved and to propose the use of MDDC for immune restoration of crewmembers with immune suppression post space flight.		

Task Description:

Ground-based models have demonstrated adverse affects on the immune system due to stress and exposure to radiation. The biological effects of exposure to radiation in space may lead to serious health consequences for crewmembers of future space flights of long duration. The ability of immune cells such as MDDC and lymphocytes to survive exposure to radiation and the ability of the surviving cells to mediate an immune response is not well understood. During long-term space flight, astronauts may become immune compromised and develop secondary immunodeficiency and malignancies.

The mechanism by which radiation exposure encountered during space flight of long duration may affect the immune system is not yet fully appreciated. Using the crewmember's own hematopoietic progenitor cells, lymphocytes, and MDDC that were collected prior to flight could be a countermeasure approach for reconstituting the immune system post radiation exposure. Even though much is known of the effects of radiation on different cell types, little is known of the maturation and function of MDDC exposed to radiation. We propose to study the effects of radiation, in particular, proton (available at the Loma Linda University), high-energy heavy ion beams (available at the Brookhaven Nat'l Lab.) and gamma-radiation on the differentialtion, maturation, and antigen presentation capability of human MDDC.

Our hypothesis is that protracted space flight will reduce the number of dendritic cells, severely impair the immune system, and seriously attenuate immune surveillance mechanisms unless appropriate countermeasures are taken.

The work will be performed at the University of Texas M. D. Anderson Cancer Center (MDACC) by Drs. James Reuben (Immunologist), Bang-Ning Lee (Molecular Biologist), and Martin Korbling, M.D. (Director of Apheresis Center, MDACC). The University of Texas M.D. Anderson Cancer Center ranks as one of the nation's most respected and productive Comprehensive Cancer Centers and is devoted exclusively to cancer patient care, research, education and prevention. We submit this proposal to study the effects of proton-, heavy ion- and gamma-radiation on monocyte-derived dendritic cells with the following specific aims: Aim #1: To determine the effect of proton-, heavy ion- and gamma-radiation on the differentiation of myeloid precursor cells into immature dendritic cells. Aim #2: To determine the effect of proton-, heavy ion- and gamma-radiation on the ability of immature DCs to capture antigen and to express costimulatory receptors. Aim #3: To determine the effect of proton-, heavy ion- and gamma-radiation on the ability of activated mature DCs to present antigen to T cells. Aim #4: To identify countermeasures that would obviate the adverse effects of proton-, heavy ion- and gamma-radiation on myelosuppression and the subsequent immune suppression. To determine the effects of low-dose radiation on the differentiation and functon of antigen presenting cells, dendritioc cells.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Project Title: Effect of Radiation on Human Peripheral Blood Dendritic Cells
Project Number: NAG 9-1454

Principal Investigator: James M. Reuben, Ph.D.

Co-Principal Investigator: Bang-Ning Lee, Ph.D.

Total Duration of Funding: 3 Years

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Institution: University of Texas M.D. Anderson Cancer Center

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Introduction

Exposure of crewmembers to space-related radiation can be a serious impediment to future space travel of long duration. Ground-based models have demonstrated adverse affects on the health of astronauts due to stress from zero gravity and exposure to radiation. Both weightlessness and high-energy ionizing radiation (HZE) is thought to have severe health implications for astronauts on protracted space flight to Mars. Space or solar radiation is acknowledged to comprise photons (X-rays), electrons (gamma-rays), protons, neutrons, and heavy metal ions (7). Even though the dose rate of HZE in space would be low, little is known of the bystander effect of HZE even at low dose rates. In a series of experiments conducted at Loma Linda University (LLU), investigators carefully delineated the effects of proton radiation upon immune structure and function in murine models (3). C57BL/6 mice were given 3 Gray (Gy) protons in place of gamma rays in one dose; some animals also received an immunization with sheep red blood cells (SRBC). By days 4 to 10 after irradiation, statistically significant decreases were found in CD19+ B cells, CD3+ T cells, CD4+ T cells, and CD8+ T cells (4). Irradiated animals showed a delayed and lowered anti-SRBC antibody response. Natural killer cells were relatively radioresistant. B cells recovered by day 15 and CD3+ and CD4+ T cells by day 29, but CD8+ T cells remained impaired. These authors also discovered that the onventional gamma radiation yielded similar data. The immunosuppressive cytokine, transforming growth factor-b1 (TGF-b1), was increased by day 7 in gamma-irradiated mice, but then fell in both proton- and gamma-ray treated animals by day 17 (3). In a paired series of experiments in this murine model system, the LLU investigators found that gamma-ray treatment acutely (4 days) decreased spleen lymphocytes more than it did peripheral blood lymphocytes (PBL); higher doses were more toxic and acutely (4 days) decreased IL-2 secretion by activated splenocytes (2). These experiments provide ample evidence that space radiation is likely, at least acutely, to affect immune forces. In this year, we have studied the effects of gamma radiation on the differentiation, maturation and function of human dendritic cells (DC), the most potent antigen presenting cells of the immune system. Because of the paucity of DC circulating in peripheral blood, we have studied the differentiation and maturation of monocyte-derived DC (MDC) in the peripheral blood of hematopoietic stem cell donors who have received hematopoietic growth factors, such as granulocyte colony stimulating factor (G-CSF). Our hypothesis is that protracted space flight will reduce the number of dendritic cells, severely impair the immune system, and seriously attenuate immune surveillance mechanisms unless appropriate countermeasures are taken. In the second year of this grant, we studied the effects of g-radiation and microgravity on monocyte-derived dendritic cells with the following specific aims: Aim #1: To determine the effect of g-radiation on the differentiation of myeloid-derived DC precursors into immature dendritic cells, their ability to present antigen to autologous T cells. Aim #2: To determine the effect of g-radiation on the gene expression of myeloid-derived DC Aim #3: To determine the combined effect of

microgravity and g-radiation on the ability of activated mature DCs to present antigen to T cells.

Critical Path Roadmap (CPR) CPR Risk(s) Addressed: Risk #22: The effects of space-based radiation will adversely effect the differentiation, maturation and function of the most potent antigen-presenting cells, the dendritic cells, and lead to the development of immune deficiency, infection, and malignancy.

Critical Question #7.03: Do factors associated with flight (stress, environment, micro-gravity, nutritional status, radiation) affect humoral or cell mediated immune function, non-specific immunity, mucosal immunity, or immune surveillance capabilities of crewmembers in a manner that exposes them to unacceptable medical risk (disease, allergy, delayed wound healing)?

In the current year of study, we hypothesized that radiation would adversely impact the differentiation, maturation, and function of dendritic cells. We used hematopoietic growth factors to mobilize immune cells, in particular, isolate monocytes and culture them in vitro to obtain monocytes-derived dendritic cells. We studied the effect of radiation and microgravity on the differentiation and maturation of monocyte-derived dendritic cells and their ability to present antigen to T lymphocytes.

Brief summary

Low dose gamma irradiation adversely affected the differentiation of peripheral blood monocytes cultured in the presence of GM-CSF and IL-4 into immature MDC and their subsequent maturation into mature MDC. Specifically, low dose gamma radiation significantly: a) increased apoptosis of immature MDC; b) decreased the ability of immature MDC to mature MDC following activation; c) deceased the expression of costimulatory molecules, HLA-DR, CD80 and CD86, on mature MDC; and d) increased the expression of Fas (CD95) on mature MDC. Collectively, these results suggest that low dose g-irradiation will compromise the ability of the host to successfully mount an immune response against pathogens.

In this year of study, we have accomplished the following: 1. Demonstrated the effects of low dose g-irradiation on the differentiation of monocytes into dendritic cells. An oral presentation of our results was made at the 25th International Gravitational Physiology Conference in Moscow, Russia in June 2004. A manuscript of our work/presentation was submitted for publication. 2. Initiated studies to investigate the effect of low dose g-irradiation on gene expression by monocytes derived dendritic cells. 3. Initiated studies to investigate the combined effects of low-dose gamma radiation and microgravity on the differentiation and maturation of monocyte derived dendritic cells.

Materials and Methods:

Peripheral blood mononuclear cell collection Peripheral blood mononuclear cells (PBMC) were prepared from all normal donors by Ficoll (Sigma Chemical Co.) density gradient centrifugation of whole blood. PBMC were washed twice with PBS and resuspended in RPMI 1640 (Biowhittaker, Walkersville, MD) supplemented with 10% heat-inactivated fetal bovine serum (J.R. Scientific, Woodland, CA), 100 U/ml of penicillin (Whittaker Bioproducts, Walkersville, MD), 100 mg/ml of streptomycin (Whittaker Bioproducts) and 2 mM of glutamine (GIBCO, Grand Island, NY).

Monocyte isolation and culturing of MDC PBMC were reacted with anti-CD14 coated magnetic beads and CD14+ monocytes were isolated by positive selection using the AutoMACS system (Miltenyi Biotec, Auburn, CA). To differentiate the CD14+ monocytes into monocyte-derived dendritic cells (MDC), 20×106 CD14+ cells suspended in 10 ml of culture medium supplemented with 1000 U/ml of GM-CSF and 500 U/ml of IL-4 (DC medium) were dispensed into T75 flasks and incubated for 6 days at 37 °C in a humidified atmosphere containing 5% CO2. On days 2 through day 5 of culture, 5 ml of fresh DC medium was added daily to the culture. On the sixth day of culture, the maturation of MDC was accomplished by incubating the immature MDC with 10 pg/ml of bacterial lipopolysaccharide (LPS) for 24 hours. On the seventh day, mature MDC were harvested.

Expression of costimulatory molecules by mature MDC To determine the level of maturation among the MDC, 2.5 x 105 MDC were suspended in 100 ml of culture medium and reacted with a panel of fluorescinated monoclonal antibody directed against CD83, CD80, CD86, and HLA-DR for 20 minutes. Appropriate Ig isotype controls were used at the same protein concentration as the test antibody to eliminate nonspecific binding of the antibodies to the MDC. Stained cells were analyzed by a flow cytometer.

Combined effects of microgravity and radiation on the differentially expressed MDC genes We cultured PBM with 1000 U/mL GM-CSF and 500 U/mL of IL-4 (DC medium) for 7 days in T-75 tissue culture flasks or 50 ml Rotating Cell Culture System (RCCS) to permit the differentiation of CD14+ PBM into immature monocyte-derived dendritic cells (iMDC). On day 7, the iMDC were harvested, counted and distributed to either T-25 flasks or 10 ml RCCS bioreactors to be activated with cytokine cocktail for an additional period up to 48 h. We studied the expression of genes in iMDC isolated under 4 experimental conditions: g-irradiated T-75 (A1), nonirradiated T-75 (A2), g-irradiated RCCS (B1), and nonirradiated RCCS (B2). Briefly, CD14+ PBM were dispensed for four separate conditions (A1, A2, B1, B2) and cultured in DC medium for 7 days to allow for their differentiation into iMDC. For each treatment, duplicate cultures were set up on day 1, iMDC culture from each treatment will be harvested on day 7. On day 7, cytokine cocktail was added to each culture to activate the iMDC to become mature MDC. On day 9, the MDC cultures from each treatment were harvested. RNA was extracted from the day 7 and day 9 cultures for hybridization to cDNA microarrays.

Gene expression profile of mature MDC RNA was extracted from 107 MDC using the “RNeasy mini kit” (Qiagen, Valencia, CA). RNA will be dissolved in RNase-Free H2O (DEPC-treated and autoclaved) at a concentration of at least 0.5 mg/ml. The A260:A280 ratio of the RNA should be at least 1.8. Characterization of total RNA samples by agarose gel electrophoresis should yield sharp (not smeared) 28S and 18S rRNA bands having an intensity ratio of 2:1. The total RNA (10 mg) were converted to cDNA with the AmpoLabeling-LPR Kit (SuperArray BioScience Corp.) that increased the limit of detection and allowed us to see transcripts you might ordinarily have missed no matter how much RNA was used. The samples were used for CG12 microarray (generated by Institutional Core Lab, array information available from http://www.mdanderson.org/ with “Radiation” labeled in the Cy5 (A, or red) channel and “Control” labeled in the Cy3 (B, or green) channel.

The combined effects of microgravity and radiation on the expression of surface antigens/receptors and production of cytokines/chemokines by MDC We have studied the expression of genes in iMDC obtained under 4 experimental conditions: irradiated T-75 (A1), nonirradiated T-75 (A2), irradiated RCCS (B1), and nonirradiated RCCS (B2). For each treatment, duplicate cultures were set up on day 1, iMDC culture from each treatment were harvested on day 7 and

Task Progress:

then activated with 50 ng sCD40L (BioSource International) and incubated for an additional 48 hours. On day 9, the mature MDC from each treatment were harvested, the cells concentrated by centrifugation and suspended in PBS plus human AB serum and used for immunophenotying of surface antigens/receptors. At various times throughout the 48-hour incubation period for the activation of iMDC with cytokine cocktail, a 200 mL aliquot of culture supernatant was withdrawn at 2h, 4h, 8h, 12h, 16h, 24h, 36h, and 48h of the activation incubation period to study the kinetics of cytokine/chemokine production. Culture supernatant was harvested and stored frozen at -20 °C for batch analysis to detect cytokines/chemokines by Luminex Multiplex Bead Array assay.

Expression of surface molecules on immature and mature MDC Immature and mature MDC were grown in T-75 flasks (A1 and A2) or in RCCS bioreactors (B1 and B2). On day 7 and day 9, the immunophenotype of iMDC from control (A2 and B2) and g-irradiated (A1 and B1) cultures was determined by flow cytometry. The immunophenotype panel included reagents to detect surface expression of DC markers (Lineage negative cocktail, CD11c, CD123, HLA-DR), costimulatory molecules (CD80, CD86), chemokine receptors (CCR5, CCR7), adhesion molecules (ICAM), and maturation marker (CD40 and CD83). MDC at 2.5 x 105 was pelleted and suspended in 100 mL of PBS with 10% human AB serum. These cells were then be stained for 20 minutes on ice using individual antibodies conjugated with one of four fluorochromes (FITC, PE, PerCP, or APC) to detect expression of markers on lineage-negative leukocytes (Lineage specific cocktail of reagents): CD11c, HLA-DR, CD86, CD80, CCR5, CCR7, ICAM, CD40 and CD83. Cells were then washed twice in ice-cold PBS, fixed with 1% paraformaldehyde and analyzed within 2 hours on a flow cytometer (FACSCalibur, Becton-Dickinson, Mountain View, CA). Appropriate Ig isotype controls were used at the same protein concentration as the test antibody to assess nonspecific reactivity. Dendritic cells were gated based on their lack of expression of lineage-specific (Lin-negative) markers and for expression of HLA-DR. Specific markers of interest were determined on DC that coexpress CD11c and each of the specific markers, i.e., coexpression of CD11c and CD80, etc. Percentage and mean channel fluorescence for each marker were recorded to evaluate the proportion of DC expressing the marker as well as the intensity (antigen density) of specific marker. Because we were staining cells with as many as four monoclonal antibodies conjugated with a different fluorochrome, we had the opportunity to examine the co-expression of up to 4 markers per cell using a flow cytometer.

Cytokines and chemokines produced by immature and mature MDC At 2h, 4h, 8h, 12h, 16h, 24h, 36h, and 48h during the in vitro activation of iMDC (Biosource International), 200 ml culture supernatants were harvested and frozen at -20 °C. At a later time, supernatants were thawed and subjected to batch analysis to assay for the presence of cytokines/chemokines by ELISA (BD-Pharmingen, San Diego, CA) or Luminex Multiplex Bead Array assay. Cytokines and chemokines were measured and included a partial list of the cytokines and chemokines produced by MDC, IL-1b, IL-6, IL-10, IL-12, IL-15, TNF-a, MIP-1b, MIP-1a, IL-8, and MCP-1 (8).

Presentation of antigen to T-cells by MDC MDC (1 x 104) were incubated with optimum concentration of recall microbial antigens (2.5 mg/ uL Tetanus toxoid or 20 mg/mL Candida albicans) or neoantigen (50 mg/mL keyhole limpet hemocyanin; KLH) in a 96-well plate for 2 hours prior to the addition of 105 autologous naïve or memory T cells and incubated for 5 days at 37 °C in a humidified atmosphere. On day 5 of culture, 1.0 mCi of 3H-thymidine (specific activity, 6.7 Ci/mmol; Dupont-New England Nuclear) in RPMI was added to each well. After incubation of the culture plate for an additional 16 h at 37 °C, the contents of the wells were harvested onto glass fiber filters and counts per minute (cpm) were determined in a liquid scintillation counter (Topcount, Beckman Coulter, Fullerton, CA). The level of DNA synthesis were reported as counts per minute (cpm) and as a stimulation index (SI) defined as a quotient of the proliferation of T cells stimulated by MDC pulsed with antigen divided by the proliferation of T cells alone.

In addition to lymphocyte blastogenesis, we examined the amount of cytokine production in these cultures. A set of cultures was set up as for lymphocyte blastogenesis and at 48 h, 200 mL of culture supernatant were harvested and stored frozen for batch analysis of cytokines. We measured IL-2, IL-4, IL-10, IFN-g, and TNF-a by Luminex Multiplex Bead Array assay.

Statistical Analysis Statistical differences between irradiated and nonirradiated treatments with respect to the capability of DC to express surface marker (CD80, CD86, CD83, CD40, etc.), cytokine levels, and the ability to present antigens to T cells were assessed by the Mann-Whitney test and the analysis of variance (ANOVA).

Results and Discussion:

The Effect of g-irradiation on MDC To investigate the effect of low-dose (LD) g-irradiation on the differentiation and maturation of MDC, peripheral blood monocytes (PBM) were isolated using magnetic bead separation technology from the peripheral blood of normal peripheral blood stem cell donors. These donors received a subcutaneous administration of granulocyte colony-stimulatory factor (G-CSF) at a dose of 6 mg/kg body weight every 12 hours on the previous 5 consecutive days to induce the recruitment of CD34+ progenitor cells into the peripheral circulation. We used these donors over that of buffy coats from normal donors, because in vitro differentiation and maturation studies of monocyte-derived dendritic cells require large numbers of CD14+ peripheral blood monocytes that were readily available from such donors. Thereafter, the CD14+ PBM were exposed to g-irradiation in doses of 5 Gy, 10 Gy, or 15 Gy, during the in vitro differentiation and maturation phases of MDC. The CD14+ PBM cells were divided into 2 groups and cultured in vitro with 1000 U/mL granulocyte macrophage colony-stimulatory factor (GM-CSF) and 500 U/mL of interleukin IL-4 (DC medium) for 7 days to permit the differentiation of CD14+ PBM into immature monocyte derived dendritic cells (iMDC). Immature MDC were cultured in medium supplemented with DC activation cocktail containing a mixture of IL-1b, IL-6, TNF-a, and prostaglandin E2 (PGE2) on day (d) 7, and the culture incubated for an additional for 2 days; all cultures were harvested on d9. The effects of g-irradiation on the expression of costimulatory molecules CD80 and CD86, Fas/CD95, and class II MHC HLA-DR were evaluated by flow cytometry. (A poster of these data was presented at the 15th Annual NASA Space Radiation Health Investigators Workshop at Port Jefferson, NY, May 16-20, 2004).

The activation of immature MDC into mature MDC Immature MDC that were exposed to 5 Gy g-irradiation on day 7 and then activated with a cytokine cocktail had a significantly lower mean channel fluorescence (MCF) of the activation marker HLA-DR (202 versus 106, P=0.027) and the costimulatory markers, CD86 (176 versus 82, P=0.036) and CD80 (167 versus 87, P=0.027), compared with nonirradiated iMDC that were activated in the same manner. The MCF of Fas/CD95 (45 versus 27, P=0.045) was also significantly decreased on mMDC that were generated from iMDC exposed to g-irradiation on d7, compared with MDC that were generated from nonirradiated iMDC. The MCF is a measure of the density of receptors on a cell and can be used to study relative levels of expression of surface antigens. These preliminary data suggest that the expressions of costimulatory receptors CD80 and CD86 on iMDC were sensitive to g-irradiation and further suggest that lower density expression of costimulatory molecules may lead to potential deficits

in their ability of iMDC to present antigen.

The proliferation of autologous T cells stimulated by mature MDCs Function of the mMDC that were differentiated from nonirradiated or g-irradiated (5 Gy) CD14+ PBM was evaluated for their ability to stimulate autologous CD3+ T cells in a mixed leukocyte reaction (MLR). The MLR cultures, consisting of quadruplicate wells of 104 MDC derived from irradiated and nonirradiated CD14+ PBM and 105 autologous T cells (final ratio of MDC to autologous T cells was 1:10) were incubated for 5 days at 37 °C in a humidified atmosphere containing 5% CO2. One mCi of 3H-thymidine was added to each well of the MLR culture for the final 16 hours of the incubation period before the cells were harvested and evaluated by liquid scintillation for the amount of radioactive thymidine incorporated into DNA synthesis. Results are expressed as a stimulation index (SI) that was a quotient of the proliferation of T cells plus MDC divided by the proliferation of T cells alone. The SI of g-irradiated (5 Gy) MDC was comparable to that of nonirradiated MDC (54 versus 51). Our preliminary data indicated that CD14+ PBM irradiated with 5 Gy of g-radiation did not affect the differentiation and maturation of MDCs, as proliferation of autologous T cells in response to activation by either irradiated or nonirradiated MDC was similar. As the stimulus of the MLR is HLA-DR on the stimulating MDC and no antigen processing is required of the MDC, this experimental design did not test the ability of MDC obtained from irradiated and nonirradiated cultures to process antigen (6). Future experiments will address the ability of irradiated MDC to process and present soluble antigens to naïve and memory T cells.

The Effect of g-irradiation on the gene expression of mature MDC We have performed a preliminary experiment comparing the gene profiles of nonirradiated mature MDC and mature MDC that were expose to 5 Gy of g-radiation. RNA was extracted from control and irradiated mMDC and subjected to the cDNA synthesis followed by hybridization to CG12 microarray, generated by M.D. Anderson Cancer Center Core Laboratory (Institutional Core Laboratory, http://www.mdanderson.org/ The samples that were irradiated (Radiation) were labeled in the Cy5 (A, or red) channel and the nonirradiated sample (Control) was labeled in the Cy3 (B, or green) channel. A cutoff of 2.2 for a significant difference between Radiation and Control samples were arbitrarily established using smooth t-statistics (S. T value). This cut-off may have been somewhat generous from a statistical viewpoint (corresponding to an unadjusted p-value of 0.027 which does not account for multiple testing). Nevertheless, using this cutoff, we were able to identify 30 genes that were significantly differentially expressed. Of these, 25 were over-expressed in the B channel (control) and 5 were over expressed in the A channel (radiation). Among the top 10 genes, the CD40 (TNFRSF5) gene expression was over expressed in nonirradiated control group; conversely stated, the expression of the CD40 (TNFRSF5) gene was reduced in MDC that were obtained from irradiated cultures, compared with those of Control cultures. This is of particular interest to our objectives as we propose to test the hypothesize that the deficits in immune function observed in immune cells during space flight are due to adverse effects of radiation (and possibly, microgravity) on the signaling through CD40 for DC activation.

Effect of simulated microgravity and g-irradiation on MDC To investigate the effect of simulated microgravity and low-dose (LD) g-irradiation on the differentiation and maturation of MDC, PBM were isolated and exposed to 3Gy g-irradiation prior to culturing the MDC in the RCCS bioreactor (Synthecon, Inc., Houston, TX). We have previously reported on the isolation of CD14+ PBM using magnetic bead separation technology from the peripheral blood of normal peripheral blood stem cell donors who had received a subcutaneous administration of G-CSF at a dose of 6 mg/kg body weight every 12 hours on the previous 5 consecutive days (9). The CD14+ PBM cells were divided into 4 aliquots; 2 aliquots received 3 Gy of g-radiation prior to incubation with 1000 U/mL GM-CSF and 500 U/mL of IL-4 (DC medium) for 7 days in the T-75 tissue culture flask (A1) or in the RCCS (B1) to permit the differentiation of CD14+ PBM into iMDC. The other 2 aliquots of CD14+ PBM were not irradiated and similarly cultured in T-75 flask (A2) or in the RCCS (B2). On day 7, iMDC from the 4 cultures were activated with DC activation cocktail containing a mixture of IL-1b, IL-6, TNF-a, and PGE-2 as before and the cultures incubated for an additional 2 days; all cultures were harvested on day 9. The expression of CD40 on the surface of mature MDC and the production of IL-12 in the culture supernatants of all cultures were determined.

CD40 expression by MDC Expression of CD40 by 3 Gy g-irradiated or nonirradiated MDC cultured in T-75 flasks or in the RCCS was determined by FACS analysis. The percentage of CD40+ MDC was similar for nonirradiated MDC cultured in A2 or B2 and the CD40 expression by irradiated MDC cultured in A1 or B1 was reduced (by approximately 50-60%), compared with expression of CD40 on MDC that were obtained from nonirradiated cultures. These data confirmed our earlier observation that cultures that were exposed to low dose g-irradiation produced a lower proportion of mature MDC, compared with nonirradiated cultures, irrespective of growth in 1 g or in simulated microgravity.

IL-12 production by MDC The activation of iMDC with a variety of agents including the agonist for CD40 leads to the production of IL-12 by mature MDC (1, 5, 10, 11). IL-12 production by mMDC obtained from cultures that were irradiated or nonirradiated and grown in T-75 flasks (A1, A2) or in the RCCS bioreactors (B1, B2) were assessed. The level of IL-12 in culture supernatants was measured using the Luminex Multiplex Bead Array assay. Compared with the production of IL-12 from the T-75 flask (A2), the production of IL-12 from the RCCS bioreactor culture (B2) was higher. However, g-irradiation reduced the level of IL-12 production (by approximately 20-30%), regardless of whether the MDC were harvested from T-75 (A1) or RCCS bioreactor (B1) cultures. Once again, these preliminary data suggest that irradiation of iMDC prior to activation yields fewer activated MDC, irrespective of whether the MDC were cultured in 1 g or microgravity simulated conditions.

The production of Type 1 cytokines by T cells activated by autologous MDC pulsed with recall antigen (Candida albicans) or neoantigen (Keyhole Limpet Hemocyanin)

To investigate the antigen presenting function of MDC, 10,000 MDC from each culture condition were pulsed with 20 mg/mL Candida albicans (recall antigen) or 50 mg/mL keyhole limpet hemocyanin (KLH; neoantigen) for 2 hours. Next, the antigen-pulsed MDCs were co-incubated with 100,000 autologous T cells at 37 °C for an additional 2 days before the culture supernatants were collected and the levels of IL-2, IFN-g and TNF-a were measured by the Luminex Multiplex Bead Array assay. The levels (pg/ml) of IL-2 (1402 versus 490), IFN-g (1319 versus 598), and TNF-a (86 versus 43) produced by T cells activated with MDC harvested from nonirradiated cultures and grown in the RCCS bioreactor were higher than T cells that were activated with nonirradiated MDC harvested from T-75 flasks. Ligation of CD40 on MDC is a major impetus for the production of these cytokines (1, 5, 10, 11). Hence, we hypothesize that the increase in production of these cytokines by T cells may be due to the higher expression of CD40 on RCCS MDC and the production of IL-12 upon ligation of CD40. 3 Gy g-irradiated MDC cultured in RCCS bioreactor (B1) had lower levels of IL-2 (658 versus 1402) and IFN-g (961 versus 1319) production compared with nonirradiated MDC cultured in RCCS (B2). Similarly, this could be due to the lower levels of CD40 expression and IL-12 production by 3 Gy g-irradiated MDC cultured in RCCS.

There were higher levels of IFN-g (1278 versus 322) and TNF-a (125 versus 94) production by T cells activated with nonirradiated MDC cultured in the RCCS, compared with nonirradiated MDC cultured in T-75 flask. As previously seen with T cells that were activated with irradiated MDC pulsed with Candida albicans, the levels of IFN-g (280 versus 1278) and TNF-a (56 versus 125) were lower in control cultures of MDC compared with MDC cultured in the RCCS bioreactor. IL-2 production by T cells was similar for all four cultural conditions even though a reduction was seen in irradiated RCCS cultures compared with control cultures (125 versus 334).

Future studies:

1. The combined effects of microgravity and g-irradiation on the expression of surface antigens/receptors and production of cytokines/chemokines by MDC upon CD40-CD40L signaling We will study the expression of genes in iMDC obtained under 4 experimental conditions: irradiated T-75 (A1), nonirradiated T-75 (A2), irradiated RCCS (B1), and nonirradiated RCCS (B2). For each treatment, duplicate cultures will be set up on day 1, iMDC culture from each treatment will be harvested on day 7 and then activated with 50 ng sCD40L (Biosource International) and incubated for an additional 48 hours. On day 9, the mature MDC from each treatment will be harvested, the cells concentrated by centrifugation and suspended in PBS plus human AB serum and used for immunophenotying of surface antigens/receptors. At various times throughout the 48-hour incubation period for the activation of iMDC with sCD40L, a 200 mL aliquot of culture supernatant will be withdrawn at 2h, 4h, 8h, 12h, 16h, 24h, 36h, and 48h of the activation period to study the kinetics of cytokine/chemokine production. Culture supernatant will be harvested and stored frozen at -20 °C for batch analysis to detect cytokines/chemokines by Luminex Multiplex Bead Array assay.

2. The combined effects of microgravity and g-irradiation on the activation of naïve and memory T cell by sCD40L activated MDC We will assess the function of MDC that were generated from nonirradiated or irradiated (at day 7 or day 9) PBM and grown in either T-75 flasks or RCCS. On day 9, cultures will be harvested and MDC will be used to test their ability to present neoantigen and recall antigens to autologous naïve and memory T cells.

3. The effects of proton and heavy metal ion irradiation on the activation of naïve and memory T cell by sCD40L activated MDC pulsed with neoantigen and recall antigens We will assess the function of MDC that were generated from nonirradiated or proton (at Loma Linda University)/ heavy-ion (at Brook Haven National Laboratory) irradiated (at day 7 or day 9) PBM and grown in T-75 flasks. On day 9, cultures will be harvested and MDC will be used to test their ability to present neoantigen and recall antigens to autologous naïve and memory T cells.

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Bibliography Type:	Description: (Last Updated: 08/05/2020)
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